Depth Camera Configuration

1. Depth Camera Configuration

Note:

① This document uses Ubuntu 20.04 and ROS2 Foxy as examples for installation. The system and ROS versions should not be lower than Ubuntu 20.04 and ROS2 Foxy, as software versions below these are outdated. Using outdated software means you will not receive security and maintenance updates, and compatibility issues will pose significant challenges, potentially leading to unsolvable problems.

② Before starting the configuration, please install virtual machine software and import Ubuntu 20.04. Refer to "2.Linux Basic Lesson".

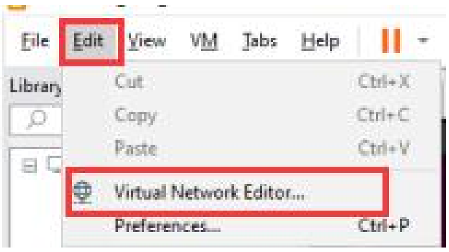
③ Next, install ROS Foxy. Refer to "Test and Configure ROS2" in the same path as this document. This document will not provide further details on the aforementioned installations.

2. Network Configuration

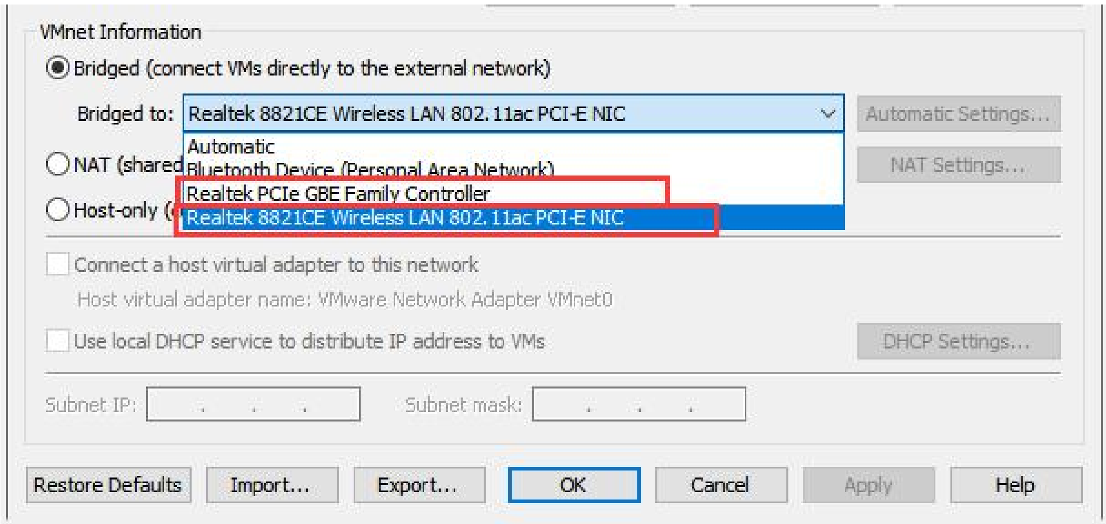
1) Open virtual machine 

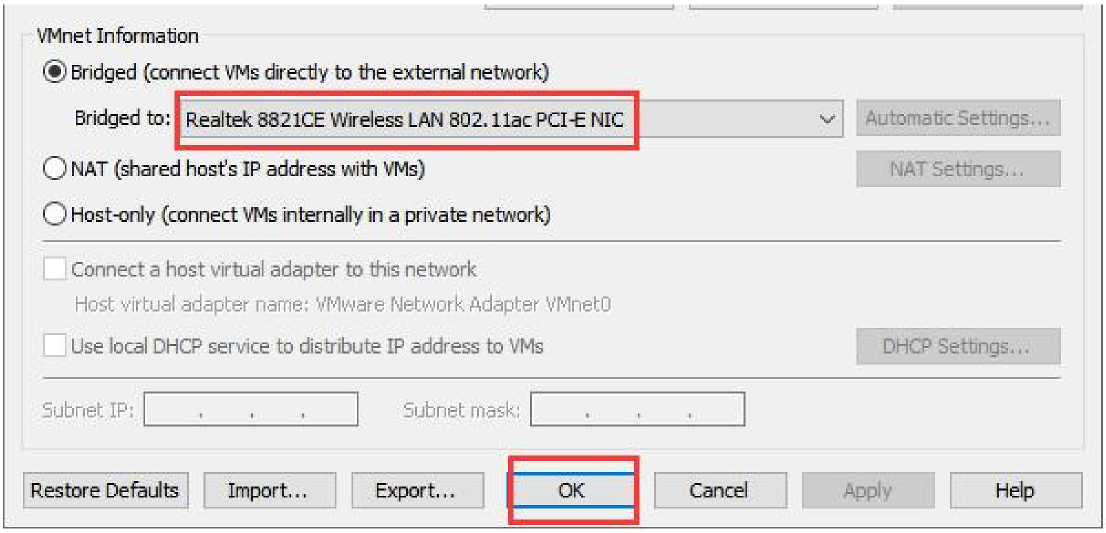
.

2) Click-on “Edit” and choose “Virtual Network Editor”.

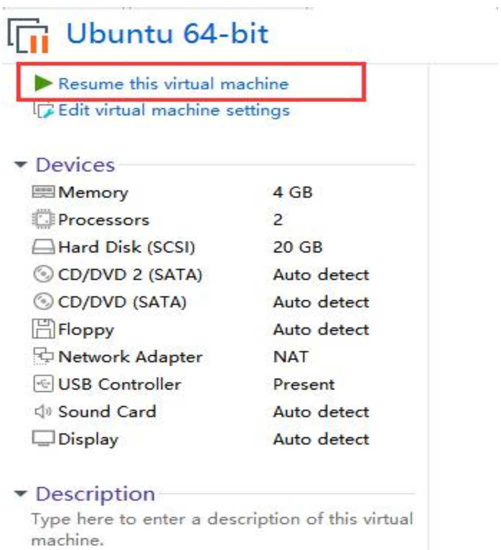


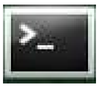
3) Tick “Bridged”, and choose the corresponding wireless network card.





4) Select the imported virtual machine, then click on “Power on this virtual machine”.



5) Click on  to open terminal. Or press the short-cut “Ctrl+Alt+t” to open the command-line terminal.

6) Enter the command to check whether the network connection is successful.

```
ping www.baidu.com
```

```
hiwonder@ubuntu:~$ ping www.baidu.com
```

If the terminal prints following content, that means network connection is successful.

```
hiwonder@ubuntu:~$ ping www.baidu.com
PING www.a.shifen.com (14.119.104.254) 56(84) bytes of data.
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=1 ttl=128 time=7.04 ms
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=2 ttl=128 time=7.28 ms
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=3 ttl=128 time=7.53 ms
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=4 ttl=128 time=7.16 ms
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=5 ttl=128 time=7.33 ms
```

If it shows that “Name or service not known”, that means virtual machine fails to connect to the network.

```
ping: www.baidu.com: Name or service not known
```

7) Execute this step only when network connection is failed. Enter command “ip a” to check the ID of network card.

```
hiwonder@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127        cope host lo
       valid        er preferred_lft forever
    inet6 :         e host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:0c:8c:ad brd ff:ff:ff:ff:ff:ff
    inet 192.168.25.129/24 brd 192.168.25.255 scope global dynamic noprefixroute
 ens33
       valid_lft 1157sec preferred_lft 1157sec
    inet6 fe80::df5b:a4ed:ab0f:b4d1/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Network card ID

8) Execute this step only when network connection is failed. Enter command and press Enter. “ens33” is the ID of network card. You need to modify the entered command based to suit the specific case at hand.

```
sudo dhclient ens33
```

```
hiwonder@ubuntu:~$ sudo dhclient ens33
```

Repeat step 6. If the terminal prints following content, that means network connection is successful.

```
hiwonder@ubuntu:~$ ping www.baidu.com
PING www.a.shifen.com (14.119.104.254) 56(84) bytes of data.
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=1 ttl=128 time=7.04 ms
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=2 ttl=128 time=7.28 ms
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=3 ttl=128 time=7.53 ms
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=4 ttl=128 time=7.16 ms
64 bytes from 14.119.104.254 (14.119.104.254): icmp_seq=5 ttl=128 time=7.33 ms
```

3. Install Dependency

1) Run the command "sudo apt update" to update apt library.

```
hiwonder@ubuntu:~$ sudo apt update
```

After update, the following messages will occur.

```
Fetched 4,435 kB in 19s (239 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

2) Enter the following commands to install the dependency library:

sudo apt install libgflags-dev nlohmann-json3-dev

**libgoogle-glog-dev **

ros-$ROS_DISTRO-image-transport

ros-$ROS_DISTRO-image-publisher

ros-$ROS_DISTRO-camera-info-manager

```
hiwonder@ubuntu:~$ sudo apt install libgflags-dev nlohmann-json3-dev libgoogle-glog-dev \
> ros-$ROS_DISTRO-image-transport ros-$ROS_DISTRO-image-publisher ros-$ROS_DISTRO-camera-info-manager
```

4. Construct Workspace

1) Execute the command to build the workspace.

mkdir -p ~/ros_ws/src && cd ~/ros_ws/src

```
hiwonder@ubuntu:~$ mkdir -p ~/ros_ws/src && cd ~/ros_ws
```

2) Execute the command to compile the new workspace. Ensure that the compilation is performed within the "ros_ws" workspace.

colcon build

```
hiwonder@ubuntu:~/ros_ws$ colcon build
```

3) Run the command to set a new workspace.

. ./install/setup.bash

```
hiwonder@ubuntu:~/ros_ws$ . ./install/setup.bash
```

4) Enter the command to append the initialization action of environment variables to the terminal initialization file.

echo "source $HOME/ros_ws/install/setup.bash" >> ~/.bashrc

```
hiwonder@ubuntu:~/ros_ws$ echo "source $HOME/ros_ws/install/setup.bash" >> ~/.bashrc
```